

Lecture Notes

EC226 Term 1 2025/26

Heteroscedasticity

Readings:

- Stock and Watson (2003): Chapter 5.4 + 9.2
- Dougherty (2016): Chapter 7
- Wooldridge (2013): Chapter 8

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1 Properties of Estimators

In the 2-variable regression model

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i \quad (1)$$

We assume $E(\varepsilon_i | \mathbf{x}) = \mathbf{0}$, $\mathbf{V}(\varepsilon_i | \mathbf{x}) = \sigma_i^2$, $\mathbf{E}(\varepsilon_i, \varepsilon_j | \mathbf{x}) = \mathbf{0}$ for $i \neq j$. Then Handout 1 (Section 4.2) on properties of estimators it was shown that

$$b_1 = \beta_1 + \frac{\sum_i (x_i - \bar{x})(\varepsilon_i - \bar{\varepsilon})}{\sum_i (x_i - \bar{x})^2} = \beta_1 + \sum_i \omega_i \varepsilon_i, \quad \text{where } \omega_i = \frac{(x_i - \bar{x})}{\sum_i (x_i - \bar{x})^2} \quad (2)$$

1.1 Unbiasedness

Taking expectations of equation (2)

$$E(b_1 | \mathbf{x}) = \mathbf{E}(\beta_1 | \mathbf{x}) + \mathbf{E}\left(\sum_i \omega_i \varepsilon_i | \mathbf{x}\right)$$

as β_1 is a constant, $E(\beta_1 | \mathbf{x}) = \beta_1$

$$E(b_1 | \mathbf{x}) = \beta_1 + \mathbf{E}\left(\sum_i \omega_i \varepsilon_i | \mathbf{x}\right) = \beta_1 + \mathbf{E}(\omega_1(\varepsilon_1 - \bar{\varepsilon}) + \omega_2(\varepsilon_2 - \bar{\varepsilon}) + \dots + \omega_n(\varepsilon_n - \bar{\varepsilon}) | \mathbf{x})$$

As ω_i is made up of x_i , (then $E(\omega_i \varepsilon_i | \mathbf{x}_i) = \omega_i \mathbf{E}(\varepsilon_i | \mathbf{x}_i)$, consequently:

$$E(b_1) = \beta_1 + \omega_1 E(\varepsilon_1 | \mathbf{x}) + \omega_2 \mathbf{E}(\varepsilon_2 | \mathbf{x}) + \dots + \omega_n \mathbf{E}(\varepsilon_n | \mathbf{x})$$

as $E(\varepsilon_i | \mathbf{x}) = \mathbf{0} \forall i$. Therefore the slope coefficient is an unbiased estimator, that is, $E(b_1 | \mathbf{x}) = \beta_1$.

1.2 Variances

Using equation (2) again, we have that

$$V(b_1 | \mathbf{x}) = E[(b_1 - E(b_1 | \mathbf{x}))^2] = E[(b_1 - \beta_1 | \mathbf{x})^2] = E\left[\left(\sum_{i=1} \omega_i \varepsilon_i | \mathbf{x}\right)^2\right]$$

$$\begin{aligned} V(b_1 | \mathbf{x}) &= E[(\omega_1 \varepsilon_1 + \omega_2 \varepsilon_2 + \dots + \omega_n \varepsilon_n | \mathbf{x})^2] = E(\omega_1^2 \varepsilon_1^2 + \omega_2^2 \varepsilon_2^2 + \dots + \omega_n^2 \varepsilon_n^2 \\ &\quad + 2\omega_1 \omega_2 \varepsilon_1 \varepsilon_2 + 2\omega_1 \omega_3 \varepsilon_1 \varepsilon_3 + \dots + 2\omega_1 \omega_n \varepsilon_1 \varepsilon_n \\ &\quad + 2\omega_2 \omega_3 \varepsilon_2 \varepsilon_3 + \dots + 2\omega_2 \omega_n \varepsilon_2 \varepsilon_n \\ &\quad + \dots + 2\omega_{n-1} \omega_n \varepsilon_{n-1} \varepsilon_n | \mathbf{x}) \end{aligned}$$

As ω_i is made up of x_i , (then $E(\omega_i^2 \varepsilon_i^2 | \mathbf{x}) = \omega_i^2 E(\varepsilon_i^2 | \mathbf{x})$, consequently:

$$\begin{aligned} V(b_1 | \mathbf{x}) &= \omega_1^2 E(\varepsilon_1^2 | \mathbf{x}) + \omega_2^2 E(\varepsilon_2^2 | \mathbf{x}) + \dots + \omega_n^2 E(\varepsilon_n^2 | \mathbf{x}) \\ &\quad + 2\omega_1 \omega_2 \underbrace{E(\varepsilon_1 \varepsilon_2 | \mathbf{x})}_0 + 2\omega_1 \omega_3 \underbrace{E(\varepsilon_1 \varepsilon_3 | \mathbf{x})}_0 + \dots + 2\omega_1 \omega_n \underbrace{E(\varepsilon_1 \varepsilon_n | \mathbf{x})}_0 \\ &\quad + 2\omega_2 \omega_3 \underbrace{E(\varepsilon_2 \varepsilon_3 | \mathbf{x})}_0 + \dots + 2\omega_2 \omega_n \underbrace{E(\varepsilon_2 \varepsilon_n | \mathbf{x})}_0 \dots + 2\omega_{n-1} \omega_n \underbrace{E(\varepsilon_{n-1} \varepsilon_n | \mathbf{x})}_0 \end{aligned}$$

As $E(\varepsilon_i \varepsilon_j | \mathbf{x}) = \text{cov}(\varepsilon_i, \varepsilon_j | \mathbf{x}) = 0$, for $i \neq j$. In addition, as $V(\varepsilon_i | \mathbf{x}) = E(\varepsilon_i^2 | \mathbf{x}) = \sigma_i^2$.

$$V(b_1 | \mathbf{x}) = \omega_1^2 \sigma_1^2 + \omega_2^2 \sigma_2^2 + \dots + \omega_n^2 \sigma_n^2 = \sum_i \omega_i^2 \sigma_i^2 = \frac{\sum_i (x_i - \bar{x})^2 \sigma_i^2}{\left[\sum_i (x_i - \bar{x})^2 \right]^2} \neq \frac{\sigma^2}{\sum_t (x_t - \bar{x})^2} \quad (3)$$

that is, the standard OLS formula (as used by computer programs, for example Stata) gives the incorrect variance estimate, consequently the t-ratios are wrong. More succinctly, we can write (3) as:

$$V(b_1) = \frac{\sigma_v^2}{\left[\sum_i (x_i - \bar{x})^2 \right]^2} \quad (4)$$

where $v_i = (x_i - \bar{x})\varepsilon_i$ and $\sigma_v^2 = \sum_i (x_i - \bar{x})^2 \varepsilon_i^2$ and this the formula for White's heteroscedastic robust standard errors, you can get Stata to estimate a model using these heteroscedastic consistent standard errors by using the option `robust` after specifying the equation (see Appendix A for an example of this).

2 Detecting Heteroscedasticity

Consider the general linear regression model

$$y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \dots + \beta_k x_{ki} + \varepsilon_i \quad (5)$$

where $V(\varepsilon_i|\mathbf{x}) = E(\varepsilon_i^2|\mathbf{x}) = \sigma_i^2$, but all of the other CLRM assumptions are satisfied. Now assuming we know the form of the heteroscedasticity as: $V(\varepsilon_i) = E(\varepsilon_i^2) = \sigma_i^2 = \delta_0 + \delta_1 z_{1i} + \dots + \delta_p z_{pi}$ in which case

$$\varepsilon_i^2 = \delta_0 + \delta_1 z_{1i} + \dots + \delta_p z_{pi} + \varsigma_i$$

where ς_i is some well-behaved error process. Under the null hypothesis of no heteroscedasticity the null hypothesis is:

$H_0 : \delta_1 = \delta_2 = \dots = \delta_p = 0 \Rightarrow \varepsilon_i^2 = \delta_0 + \varsigma_i$ (the variance of the error term depends solely on a constant - homoscedasticity).

$H_1 : \text{Any } \delta_j \neq 0$ (the variance of the error term depends on z_{ji} - heteroscedasticity).

Unfortunately, ε_i is unobserved and we use the residuals (e_i) from equation (5) as a proxy for these, in which case we have:

$$e_i^2 = \delta_0 + \delta_1 z_{1i} + \dots + \delta_p z_{pi} + \varsigma_i \quad (6)$$

although the null and alternative hypothesis remain as above and this is tested via the usual F-test. The question now arises as to the form of $z_{1i}, \dots, z_{pi}$. Stata has programmed within it 3 proxies for these variables:

1. White's Heteroscedasticity Test (without cross terms)

In this case $z_{1i} = x_{1i}, z_{2i} = x_{2i}, \dots, z_{ki} = x_{ki}, z_{k+1i} = x_{1t}^2, \dots, z_{pi} = x_{ki}^2$, in which case $p = 2k$.

After estimating an equation the Stata command to test for heteroscedasticity is:

`estat hettest list of variables`

(see Appendix A for an example of this test)

2. Breusch-Pagan test

In this case $z_{1i} = \hat{y}_i$, in which case $p = 1$ and the Stata command is

`estat hettest`

3. ARCH Test

This test for heteroscedasticity is when you have time series data, in this case $z_{1i} = e_{i-1}^2, \dots, z_{pi} = e_{i-p}^2$.

(After estimating an equation the Stata command to test for heteroscedasticity is:

`estat archlm, lags(p)`

where p is the lag you wish to include).

In all cases you are using only proxies for the true z variables, which are unknown as a result the power of this test will tend to be LOW.

BEWARE

Apparent heteroscedasticity could be caused by an omitted relevant variable. Suppose we have two models, denotes as the true model (T) and the false model (F):

$$y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \varepsilon_i \quad (T)$$

where ε_i satisfies CLRM assumptions. But you incorrectly estimate the false model

$$y_i = \delta_0 + \delta_1 x_{1i} + \eta_i \quad (\text{F})$$

where $\eta_i = \beta_2 x_{2i} + \varepsilon_i$, in which case $\text{var}(\eta_i) = E(\eta_i^2) = E[(\beta_2 x_{2i} + \varepsilon_i)^2] = \beta_2^2 E(x_{2i}^2) + E(\varepsilon_i^2)$. As ε_i satisfies the CLRM assumptions it has a constant variance, but the $\text{var}(x_{2i})$ is unlikely to be constant, in which case we have:

$$\text{var}(\eta_i) = E(\eta_i^2) = E[(\beta_2 x_{2i} + \varepsilon_i)^2] = \beta_2^2 E(x_{2i}^2) + \sigma^2$$

and we get apparent heteroscedasticity caused by an omitted relevant variable. In this case we do NOT actually have heteroscedasticity it is simply a symptom of something else.

NOTE:

If you have heteroscedasticity in the error term and you use some form of heteroscedastic standard errors then the F-tests we talked about in Handout 2 (Section 6.2) are no longer appropriate (see Appendix A).

Appendices

A Example of heteroscedastic

$$lhourpay_i = \delta_0 + \delta_1 age_i + \delta_2 age_i^2 + \delta_3 edage_i + \gamma_0 sex_i + \gamma_1 2.ethnic_i + \gamma_2 3.ethnic_i + \varepsilon_i, \quad i = 1, \dots, 9526$$

Source	SS	df	MS	Number of obs	=	9,526
Model	534.877205	6	89.1462008	F(6, 9519)	=	348.04
Residual	2438.16961	9,519	.256137158	Prob > F	=	0.0000
				R-squared	=	0.1799
				Adj R-squared	=	0.1794
Total	2973.04682	9,525	.312130899	Root MSE	=	.5061

lhourpay	Coefficient	Std. err.	t	P> t	[95% conf. interval]	
age	.0624868	.0026719	23.39	0.000	.0572493	.0677243
c.age#c.age	-.0006343	.0000304	-20.84	0.000	-.0006939	-.0005746
edage	.0536192	.0017302	30.99	0.000	.0502276	.0570109
sex						
Female	-.210526	.0103917	-20.26	0.000	-.230896	-.1901561
ethnic						
Asian	-.1410976	.0235136	-6.00	0.000	-.1871893	-.095006
Other	-.1137099	.0249926	-4.55	0.000	-.1627008	-.064719
_cons	.2533564	.0625185	4.05	0.000	.1308069	.3759059

To test the significance of the ethnic dummies we are testing: $H_0 : \gamma_1 = \gamma_2 = 0$ we have to estimate a restricted model:

Source	SS	df	MS	Number of obs	=	9,526
Model	521.22848	4	130.30712	F(4, 9521)	=	506.01
Residual	2451.81834	9,521	.257516893	Prob > F	=	0.0000
				R-squared	=	0.1753
				Adj R-squared	=	0.1750
Total	2973.04682	9,525	.312130899	Root MSE	=	.50746

lhourpay	Coefficient	Std. err.	t	P> t	[95% conf. interval]	
age	.0616826	.0026767	23.04	0.000	.0564357	.0669294
c.age#c.age	-.0006238	.0000305	-20.47	0.000	-.0006836	-.0005641
edage	.0517347	.0017154	30.16	0.000	.0483722	.0550971
sex						
Female	-.2095656	.0104187	-20.11	0.000	-.2299885	-.1891427
_cons	.2889754	.0624939	4.62	0.000	.1664741	.4114767

And the F-test is:

$$F = \frac{(2451.81834 - 2438.16961)/2}{2438.16931/9519} = 26.64$$

And we can get the same result from typing:

```
. test 2.ethnic 3.ethnic

( 1) 2.ethnic = 0
( 2) 3.ethnic = 0

F( 2, 9519) = 26.64
Prob > F = 0.0000
```

However, we see that if we test for the presence of heteroscedasticity in the error term in the unrestricted equation, then:

```
Breusch-Pagan/Cook-Weisberg test for heteroskedasticity
Assumption: i.i.d. error terms
Variable: Fitted values of lhourpay

H0: Constant variance

F(1, 9524) = 75.66
Prob > F = 0.0000
```

There is significant evidence to reject the null hypothesis that the errors are homoscedastic. If we believed our model was correctly specified then we might choose to re-estimate with heteroscedastic robust standard errors, i.e.:

```
Linear regression               Number of obs   =    9,526
                               F(6, 9519)         =   277.66
                               Prob > F           =    0.0000
                               R-squared          =    0.1799
                               Root MSE       =    .5061
```

lhourpay	Coefficient	Robust std. err.	t	P> t	[95% conf. interval]	
age	.0624868	.0027944	22.36	0.000	.0570093	.0679643
c.age#c.age	-.0006343	.000033	-19.24	0.000	-.0006989	-.0005696
edage	.0536192	.0024115	22.23	0.000	.0488922	.0583463
sex						
Female	-.210526	.0104009	-20.24	0.000	-.2309139	-.1901381
ethnic						
Asian	-.1410976	.0259298	-5.44	0.000	-.1919257	-.0902696
Other	-.1137099	.0238566	-4.77	0.000	-.160474	-.0669458
_cons	.2533564	.0683357	3.71	0.000	.1194037	.387309

But if we again test $H_0 : \gamma_1 = \gamma_2 = 0$, we get:


```
. test 2.ethnic 3.ethnic

( 1)  2.ethnic = 0
( 2)  3.ethnic = 0

F( 2, 9519) = 24.63
Prob > F = 0.0000
```

But this clearly does NOT agree with our previous F-test and that is because Stata does not use the RSS version of the formula to calculate F-tests, but rather uses the vector of coefficient estimates and the variance-covariance matrix see Handout 2: Appendix E.

B Stata Commands for heteroscedasticity

```
/* Running a regression and saving the residual and fitted values */
reg lhourpay age c.age#c.age edage i.sex i.ethnic if(edage>=0 & edage<=95 & ethnic~=.)
predict res2, residual
predict yhat2, xb
/* Stata's test for heteroscedasticity: Breusch-Pagan)
estat hettest, fstat
scalar hF1a=r(F)
/* Stata's test for heteroscedasticity: White's)
estat hettest age c.age#c.age edage i.sex i.ethnic c.age#c.age#c.age /
*/ c.edage#c.edage, fstat
scalar hF2a=r(F)
/* generating residuals squared */
gen res2sq=res2*res2
/* Breush-Pagan test for Heteroscedasticity */
reg res2sq yhat if(edage>=0 & edage<=95 & ethnic~=.)
test yhat
scalar hF1b=r(F)
/* White's test for Heteroscedasticity */
reg res2sq age c.age#c.age edage i.sex i.ethnic c.age#c.age#c.age /
*/ c.edage#c.edage if(edage>=0 & edage<=95 & ethnic~=.)
test age c.age#c.age edage 2.sex 2.ethnic 3.ethnic c.age#c.age#c.age /
*/ c.edage#c.edage
scalar hF2b=r(F)
/* Comparing Stata test with own test */
scalar list hF1a hF1b
/* Comparing Stata test with own test */
scalar list hF2a hF2b
```

C R Commands for heteroscedasticity

```
#In R you must install the appropriate package. Having installed the package
#you must load the library associated with that package each time you enter R.
#I used the following libraries over Term 1:

library(tidyverse)
library(reshape2)
library(foreign)
library(ggplot2)
library(haven)
library(readxl)
library(Hmisc)
library(ggpubr)
library(rstatix)
library(groupedstats)
library(car)
library(lmtest)
library(olsrr)
library(estimatr)
library(tseries)
library(margins)
library(ivreg)

# Running a regression and saving the residual and fitted values
model21 <- lm(lhourpay ~ age + I(age^2) + edage1 + sex + ethnic,
              data = subset(qlfs18_1, edage1>=0 & edage1<=95))
qlfs18_1$res21 <- resid(model21)
qlfs18_1$yhat21 <- fitted(model21)
# Testing significance of the ethnicity variable
test21_a <- linearHypothesis(model21, c("ethnicAsian=0", "ethnicOther=0"), test="F")
fstat21_a <- test21_a$F[2]
# Constructing a Hetero test involving yhat as F-test
htest1_a <- ols_test_f(model21)
fhstat1_a <- htest1_a$f
# Constructing a Hetero test involving all RHS variables as F-test
htest2_a <- ols_test_f(model21, rhs = T)
fhstat2_a <- htest2_a$f
# Constructing a Hetero White's test as chi-sq
htest3_a <- bptest(model21, ~ age + I(age^2) + edage1 + sex + ethnic + I(age^4) +
                  I(edage1^2), data = subset(qlfs18_1, edage1>=0 & edage1<=95))
fhstat3_a <- htest3_a$statistic
# Constructing a Hetero test involving yhat as F-test
hmodel1_b <- lm(resid(model21)^2 ~ yhat21, data = subset(qlfs18_1, edage1>=0 & edage1<=95))
hstat1_b <- linearHypothesis(hmodel1_b, c("yhat21=0"), test="F")
```

```

fhstat1_b <- htest1_b$F[2]
# Constructing a Hetero test involving all RHS variables as F-test
hmodel2_b <- lm(resid(model21)^2 ~ age + I(age^2) + edage1 + sex + ethnic,
               data = subset(qlfs18_1, edage1>=0 & edage1<=95))
htest2_b <- linearHypothesis(hmodel2_b, c("age=0", "I(age^2)=0", "edage1=0", "sexFemale=0",
                                          "ethnicAsian=0", "ethnicOther=0"), test="F")

fhstat2_b <- htest2_b$F[2]
# Constructing a Hetero White's test as chi-sq
hmodel3_b <- lm(resid(model21)^2 ~ age + I(age^2) + edage1 + sex + ethnic + I(age^4) +
               I(edage1^2), data = subset(qlfs18_1, edage1>=0 & edage1<=95))
fhstat3_b <- nobs(hmodel3_b)*summary(hmodel3_b)$r.squared
fhstat3_b <- htest3_b$Chisq[2]
# Comparing the different tests
c(fhstat1_a, fhstat1_b)
c(fhstat2_a, fhstat2_b)
c(fhstat3_a, fhstat3_b)
#Using robust standard errors and
model22 <- lm_robust(lhourpay ~ age + I(age^2) + edage1 + sex + ethnic,
                   data = subset(qlfs18_1, edage1>=0 & edage1<=95), se_type = "stata")
vb22 <- vcov(model22)
#Testing significance of ethnicity variables, note not same as the F-test using RSS
test21_b <- linearHypothesis(model22, c("ethnicAsian=0", "ethnicOther=0"),
                             test="F", vcov= vb22)
fstat21_b <- test21_b$F[2]
c(fstat21_a, fstat21_b)

```

References

- Dougherty, C. (2016). *Introduction to Econometrics*. OUP Catalogue. Oxford University Press.
- Stock, J. and Watson, M. W. (2003). *Introduction to Econometrics*. Prentice Hall, New York.
- Wooldridge, J. M. (2013). *Introduction to Econometrics*. Cengage.